

Practice Questions for Dartmouth Workshop

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1 Questions

1. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
 - a. To develop a computer program that could play chess.
 - b. To create a machine that could simulate any aspect of learning or intelligence.
 - c. To establish a standard for measuring machine intelligence.
 - d. To explore the ethical implications of artificial intelligence.
 - e. To build a robot that could perform physical tasks.
2. Which of the following best describes the Turing Test?
 - a. A test to determine if a machine can perform complex calculations.
 - b. A test to determine if a machine can exhibit intelligent behavior indistinguishable from a human.
 - c. A test to determine if a machine can learn from data.
 - d. A test to determine if a machine can solve mathematical problems.
 - e. A test to determine if a machine can perform physical tasks.
3. According to the lecture, why is brute-force search alone insufficient for complex games like Checkers or Go?
 - a. Because the rules of these games are too complex to encode.
 - b. Because the search space is too large to explore exhaustively.
 - c. Because computers are not fast enough to perform the calculations.
 - d. Because these games require human intuition and creativity.
 - e. Because the game trees are not well-defined.
4. What is a key difference between the approach used by Deep Blue and AlphaGo in playing games?
 - a. Deep Blue relied heavily on machine learning, while AlphaGo used brute-force search.
 - b. Deep Blue used brute-force search, while AlphaGo combined deep learning with search.
 - c. Deep Blue used a single model, while AlphaGo used two models.
 - d. Deep Blue was trained on self-play, while AlphaGo was trained on human games.

- e. Deep Blue was designed for Go, while AlphaGo was designed for chess.
5. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
- a. To develop a computer program that could play chess
 - b. To explore the possibility of simulating human intelligence with machines
 - c. To create a new programming language for AI research
 - d. To establish a formal definition of artificial intelligence
 - e. To build a robot that could perform simple tasks
6. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
- a. To develop a computer program that could play chess
 - b. To create a machine that could simulate any aspect of learning or intelligence
 - c. To establish a new field of study called Artificial Intelligence
 - d. To build a robot that could perform simple tasks
 - e. To prove that machines could think
7. What is the core idea behind the Turing Test?
- a. To measure a machine's ability to perform complex calculations
 - b. To determine if a machine can exhibit intelligent behavior indistinguishable from a human
 - c. To assess a machine's physical capabilities
 - d. To evaluate a machine's ability to learn from data
 - e. To test a machine's ability to follow instructions
8. According to the lecture, why is brute-force search alone insufficient for complex games like Checkers or Go?
- a. Because computers are not fast enough to evaluate all possible moves
 - b. Because the rules of these games are too complex for computers to understand
 - c. Because these games require human intuition and creativity
 - d. Because the search space is too large to explore exhaustively
 - e. Because computers cannot represent the game state accurately
9. What is a key difference between the approach used by Deep Blue and AlphaGo in playing games?
- a. Deep Blue used machine learning, while AlphaGo used brute-force search
 - b. Deep Blue relied heavily on brute-force search, while AlphaGo used deep learning and search
 - c. Deep Blue used a Monte Carlo search, while AlphaGo used a minimax algorithm
 - d. Deep Blue was trained on human games, while AlphaGo was trained on self-play

- e. Deep Blue used a single model, while AlphaGo used two models
10. According to the lecture, what was Seymour Papert's view on computer vision in 1966?
- a. It was an extremely difficult problem that would take decades to solve
 - b. It was a relatively easy problem that could be solved in a summer project
 - c. It was not a suitable problem for AI research
 - d. It was a problem that required a large amount of computational power
 - e. It was a problem that could only be solved by humans
11. What is the primary function of the first model in the AlphaGo system?
- a. To predict the likelihood of winning given the current state
 - b. To predict the next move given the current state of the board
 - c. To evaluate the game tree using Monte Carlo search
 - d. To learn from self-play games
 - e. To combine the outputs of the two models
12. According to the lecture, what is a key component of an algorithm for playing games?
- a. The ability to generate random moves
 - b. The ability to represent the state of the game
 - c. The ability to communicate with other players
 - d. The ability to learn from human players
 - e. The ability to predict the weather
13. What is the main purpose of the step function in a perceptron?
- a. To calculate the weighted sum of the inputs
 - b. To apply a bias to the inputs
 - c. To produce a binary output based on the weighted sum
 - d. To adjust the weights of the connections
 - e. To store the input values
14. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
- a. To develop a new programming language
 - b. To create a machine that could simulate any aspect of learning or intelligence
 - c. To establish a new research institute
 - d. To study the human brain
 - e. To create a new type of computer

2 Answers

1. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
 - a. While game playing is a part of AI, the Dartmouth Workshop's goal was broader than just chess.
 - b. The Dartmouth Workshop aimed to explore the possibility of machines simulating any aspect of learning or intelligence, as stated in the proposal quote on the slide.
 - c. The Dartmouth Workshop did not focus on establishing a standard for measuring machine intelligence.
 - d. The ethical implications of AI were not the primary focus of the 1956 Dartmouth Workshop.
 - e. The Dartmouth Workshop focused on cognitive abilities, not physical tasks.
2. Which of the following best describes the Turing Test?
 - a. The Turing Test is not about complex calculations, but about intelligent behavior.
 - b. The Turing Test is designed to assess if a machine can behave intelligently enough to be indistinguishable from a human in conversation.
 - c. The Turing Test does not directly test the ability to learn from data.
 - d. The Turing Test is not about solving mathematical problems, but about exhibiting human-like intelligence.
 - e. The Turing Test is not about physical tasks, but about cognitive abilities.
3. According to the lecture, why is brute-force search alone insufficient for complex games like Checkers or Go?
 - a. The rules of these games are well-defined and can be encoded.
 - b. The lecture states that the search tree sizes for games like Checkers and Go are astronomically large, making it impossible to search the entire tree even with fast computers.
 - c. While computational speed is a factor, the primary issue is the sheer size of the search space.
 - d. While human intuition is valuable, AI can learn to play these games effectively without it.
 - e. The game trees for these games are well-defined, but too large to search exhaustively.
4. What is a key difference between the approach used by Deep Blue and AlphaGo in playing games?
 - a. Deep Blue used brute-force search, not machine learning, as its primary approach.
 - b. The lecture mentions that Deep Blue's success was largely due to brute-force search, while AlphaGo combined deep learning with Monte Carlo tree search.
 - c. While AlphaGo used two models, Deep Blue's approach was not based on a single model.
 - d. AlphaGo was trained on both human games and self-play, while Deep Blue did not use self-play.
 - e. Deep Blue was designed for chess, while AlphaGo was designed for Go.
5. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?

- a. Answer A is incorrect because while game playing was a topic of interest, it was not the primary goal of the workshop.
 - b. Answer B is correct because the Dartmouth Workshop aimed to investigate whether every aspect of learning or intelligence could be simulated by a machine.
 - c. Answer C is incorrect because the workshop's focus was on the broader concept of AI, not on creating a specific programming language.
 - d. Answer D is incorrect because the workshop was more exploratory than definitional.
 - e. Answer E is incorrect because the workshop's focus was on simulating intelligence, not on building physical robots.
6. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
- a. While game playing was a topic of interest, the primary goal was broader than just chess.
 - b. The Dartmouth Workshop aimed to explore the possibility of precisely describing any aspect of learning or intelligence so that a machine could simulate it.
 - c. While the workshop is considered the birth of AI, the goal was not to establish the field, but to explore the possibility of AI.
 - d. The workshop's focus was on simulating intelligence, not building physical robots.
 - e. The workshop aimed to explore the possibility of simulating intelligence, not to prove that machines could think.
7. What is the core idea behind the Turing Test?
- a. The Turing Test is not about computational power, but about the ability to exhibit human-like behavior.
 - b. The Turing Test is designed to see if a machine can behave in a way that is indistinguishable from a human, thus demonstrating intelligent behavior.
 - c. The Turing Test is not concerned with a machine's physical capabilities, but its ability to simulate human intelligence.
 - d. While learning is a component of intelligence, the Turing Test is not specifically about learning from data.
 - e. The Turing Test is not about following instructions, but about exhibiting human-like behavior.
8. According to the lecture, why is brute-force search alone insufficient for complex games like Checkers or Go?
- a. While speed is a factor, the primary issue is the sheer size of the search space, not just the speed of computation.
 - b. Computers can understand the rules of these games; the issue is the vast number of possible moves.
 - c. While human intuition is valuable, the lecture focuses on the computational limitations of brute-force search.
 - d. The lecture emphasizes that the search space for games like Checkers and Go is astronomically large, making it impossible to explore all possibilities with brute-force search.

- e. Computers can represent the game state; the problem is the computational cost of exploring all possible states.
9. What is a key difference between the approach used by Deep Blue and AlphaGo in playing games?
- a. Deep Blue did not use machine learning as its primary approach, and AlphaGo did not use brute-force search.
 - b. Deep Blue's success was primarily due to brute-force search, while AlphaGo combined deep learning with Monte Carlo tree search.
 - c. Deep Blue did not use Monte Carlo search, and AlphaGo did not use a minimax algorithm.
 - d. AlphaGo was trained on both human games and self-play, while Deep Blue did not use machine learning.
 - e. Both Deep Blue and AlphaGo used multiple models, but the key difference is the use of brute-force search vs deep learning.
10. According to the lecture, what was Seymour Papert's view on computer vision in 1966?
- a. Papert did not think it was a difficult problem at the time.
 - b. Seymour Papert believed that computer vision was a tractable and "easy" problem that could be solved in a summer project.
 - c. Papert believed it was a suitable problem, just an easy one.
 - d. Papert did not think it required a large amount of computational power.
 - e. Papert believed it could be solved by machines.
11. What is the primary function of the first model in the AlphaGo system?
- a. The second model in AlphaGo predicts the likelihood of winning.
 - b. The first model in AlphaGo is trained to predict the next move based on the current board state.
 - c. Monte Carlo search is used to combine the two models, not the function of the first model.
 - d. The second model is trained on self-play games.
 - e. The system combines the two models, not the function of the first model.
12. According to the lecture, what is a key component of an algorithm for playing games?
- a. Generating random moves is not a key component of a good game-playing algorithm.
 - b. Representing the state of the game is a fundamental component of any game-playing algorithm.
 - c. Communication with other players is not a necessary component of a game-playing algorithm.
 - d. While learning from human players can be useful, it is not a key component of all game-playing algorithms.
 - e. Predicting the weather is not relevant to game-playing algorithms.
13. What is the main purpose of the step function in a perceptron?

- a. The weighted sum is calculated before the step function is applied.
 - b. The bias is added to the weighted sum, not applied by the step function.
 - c. The step function in a perceptron is used to produce a binary output (0 or 1) based on the weighted sum of the inputs.
 - d. The weights are adjusted during the learning process, not by the step function.
 - e. The input values are not stored by the step function.
14. According to the lecture, what was the primary goal of the 1956 Dartmouth Workshop?
- a. Answer A is incorrect because the workshop's focus was on AI, not programming languages.
 - b. Answer B is correct because the Dartmouth Workshop aimed to explore the possibility of machines simulating human intelligence.
 - c. Answer C is incorrect because the workshop was a research project, not the establishment of a new institute.
 - d. Answer D is incorrect because the workshop focused on simulating intelligence, not studying the human brain directly.
 - e. Answer E is incorrect because the workshop focused on AI, not hardware development.